

What we're building, and what we still need to decide

Status as of Thu 24 Sep 2026 Code and video due Mon 28 Sep, 08:00

Please read this before we meet. It holds our draft problem statement, how the model follows from it, the choices already made, and the **7 decisions the team still has to make**. Every part of the model has to answer part of the problem statement. If it doesn't, it doesn't go in.

1. The problem statement [TEAM TO APPROVE]

Turnout in South Africa's local government elections fell from 57.97% in 2016 to 45.86% in 2021, the lowest on record. Young people are the most disengaged group: turnout among registered voters aged 20–29 fell from 50% to 35% over the same period. Ahead of the 4 November 2026 elections, a record registration drive has pushed the voters' roll past 29 million. Online registration made up 74% of transactions, and half of all online users were aged 16–29.

The open question is whether this surge in registration will turn into votes, and where it is least likely to.

Which municipalities and voting districts across South Africa are most at risk of low turnout in 2026, how much of that risk comes from registration gaps versus turnout gaps, and where does a young voters' roll overlap with historically low turnout?

Who it's for: the IEC and civil society, to target voter education; journalists, to see which areas to watch; and parties, to see where registration isn't turning into votes.

Limits we state openly: digital-registration and age-level turnout figures are only published nationally or by province, so we use them as context and as a check, not as model inputs. Municipal youth figures are a 2026 snapshot, so there is no municipal youth trend over time.

2. How the model follows from the problem statement

The problem statement asks three things. Each one gets its own piece of analysis, and together they feed one dashboard.

"...most at risk of low turnout in 2026..."

PART A · MACHINE LEARNING

Turnout forecast

PREDICTS	PR-ballot turnout (votes cast ÷ registered voters) for each voting district. Voting-district results are then added up to municipality level, weighted by registered voters.
LEARNS FROM	Only information known before the election: turnout at the previous one and the one before, the change between them, growth in registered voters, previous spoilt-ballot rate, and Census 2022 context (internet access, basic services, hunger, urban or rural).
TESTED BY	Backtesting on real past elections: train on earlier elections, predict 2016 and 2021, and compare with what actually happened. The model must beat a simple rule, "same turnout as last time", or we don't use it.

MODELS	The "same as last time" rule, then ridge regression, then gradient boosting. We keep whichever has the lowest error (in percentage points) and ranks the lowest-turnout areas correctly.
OUTPUT	2026 turnout forecast with an error range for every voting district and municipality.

"...how much of that risk comes from registration gaps versus turnout gaps..."

PART B · CALCULATED MEASURE

Gap breakdown

MEASURES	Registration rate = registered voters ÷ adults living there. Turnout = votes ÷ registered voters. Participation = votes ÷ adults = registration rate × turnout.
ANSWERS	For each municipality: is the problem mainly that people aren't on the roll, or that people on the roll don't vote? Each needs a different response from the IEC.
NEEDS	An adult-population count per municipality. See decision 5 below.

"...where does a young voters' roll overlap with historically low turnout?"

PART C · PROFILES AND PRIORITY

Participation profiles and a priority list

PROFILES	Clustering (k-means) groups municipalities by turnout history, registration gap, youth share of the 2026 roll and Census context. We check that the groups stay the same when the settings change.
PRIORITY LIST	A transparent score combining forecast low turnout, a high youth share of the roll and a large registration gap, with the weights shown on screen.
CHECKS	National IEC figures (youth turnout 50% to 35%, 74% of registrations online) are used to motivate the question and check the results, never as model inputs.

Important: the youth share of the roll exists only for 2026, so the forecast model can't learn from it. There are no past years to train on. Youth therefore enters in Part C, not Part A. If a judge asks why the forecast doesn't use youth, this is the honest answer.

3. Already decided

DECISION	STATUS	DETAIL
Geographic scope: all 9 provinces	[DECIDED]	Other provinces' 2016 and 2021 results come from the DATASETS/L2 and L3 files, since the main baseline file only covers Mpumalanga after 2006.
Model level: voting district and municipality	[DECIDED]	Train at voting-district level (about 20,000+ districts nationally) for enough data, then add up to municipalities for the story and the dashboard.
Python environment rebuilt	[DONE]	A fresh .venv with Python 3.13, pandas, scikit-learn, openpyxl, matplotlib and Streamlit, with exact versions saved in requirements.txt. The old copy from the other computer is kept as .venv_old. In VS Code, choose the .venv kernel.

DECISION	STATUS	DETAIL
Digital registration and youth: keep them as context	[DECIDED]	47 published IEC figures are saved with source links in DATASETS/iec_2026_registration_press_figures.csv. Every row is marked as context only.

4. Decisions the team must make

Each decision lists the options, with the **recommended** one first. Please agree on all 7 so the notebook work can start.

1 · Approve the problem statement **[OPEN]**

RECOMMENDED	Approve as written It asks three questions, and each one maps directly to a part of the model.
Option	Change the wording This is fine, but any new claim needs data that supports it.

2 · What age counts as "youth"? **[OPEN]**

RECOMMENDED	18–29 The IEC roll uses the age bands 18–19, 20–29 and 30–39, so 18–29 can be counted exactly. It also matches the IEC youth figures.
Option	18–34 (South Africa's national youth definition) The roll bands can't be split at 34, so we'd have to estimate.

3 · Which past elections do we train on? **[OPEN]**

RECOMMENDED	2011, 2016 and 2021 Boundaries are mostly stable over these years, which gives two backtests (predict 2016, predict 2021). Other provinces' 2011 results appear to be in Book1.xlsx. <i>Not yet checked.</i>
Option	2016 and 2021 only Safest, but only one backtest, and each area has just one previous election to learn from.
Option	All of 2000–2021 The most history, but municipal boundaries changed heavily between 2000 and 2011, so mapping districts across years is risky.

4 · Which ballot defines turnout? **[OPEN]**

RECOMMENDED	PR ballot Every voter in a municipality gets a PR ballot, so it measures the whole municipality consistently.
Option	Ward ballot Some voters don't get one (e.g. voters outside their ward), so the counts differ.

5 · Where do adult-population numbers come from (for the registration gap)? **[OPEN]**

RECOMMENDED	Census 2022 person file (F21 in the zip) One up-to-date count per municipality for every province. <i>Not yet checked</i> whether it has age and municipality columns.
Option	IEC Atlas voting-age population file Already in DATASETS, but it stops at 2016, with nothing for 2021 or 2026.

6 · How do we deploy the model? [\[OPEN\]](#)

RECOMMENDED

Streamlit dashboard

Already installed. It supports a map or table, forecasts, profiles and a municipality drill-down, and it's easy to demo in the video.

Option

Interactive notebook only

It meets the rubric's minimum, but it looks weaker in the demo.

7 · Who does what, and who presents what? [\[OPEN\]](#)

Every member must present in the 15-minute video. Suggested split, one section per person:

- Problem and why it matters, plus data sources and cleaning
- Exploring the data and the gap breakdown (Part B)
- Forecast model and backtest results (Part A)
- Profiles, priority list and dashboard demo (Part C), plus limitations and recommendations

5. Data fixes before any modelling

PROBLEM	FIX	STATUS
The Census file was cut off at Excel's row limit (1,048,575). There are no Mpumalanga or Limpopo households.	Use the full Census2022sample F18/F19 files from DIRISA_Election_Challenge.zip.	[NEXT]
The baseline only has all 9 provinces for 2000 and 2006.	Build 2011/2016/2021 for all provinces from L2, L3 and Book1.	[NEXT]
Boundaries changed (e.g. MP322 and MP323 merged into MP326 in 2016).	Map every voting district onto 2021 municipalities.	[NEXT]
MP314 is labelled "Emlahleni" in the 2026 workbook.	It's Emakhazeni, confirmed by the 2026 municipal code list and the VAP file.	[UPDATE CROSSWALK]
The source files "MP 2016.csv" and "MP 2021.csv" had their years swapped.	Already corrected in the baseline using DateGenerated (year_label_corrected). We mention this in the write-up.	[HANDLED]
Notebook bugs: re-reads a 487 MB file, removes duplicates on a mostly-empty column, youth CSV header offset, output files loose in the project root.	Rewrite Stages 2–3 together, cell by cell.	[AFTER DATA FIXES]
Our turnout totals haven't been checked against the IEC's.	Compare with the IEC's official turnout files in the zip.	[NEXT]
The Git repository covers the whole Documents folder: CVs, games, gigabytes of data.	Make a separate repository just for the project, with the data folders excluded.	[DON'T COMMIT YET]

6. Plan to the deadline

DAY	WORK	DONE WHEN
Thu 24 Sep	Agree the 7 decisions. Fix the Census, build the all-province election panel, map boundaries, check totals against the IEC.	One clean table per voting district per election, with totals matching the IEC.

DAY	WORK	DONE WHEN
Fri 25 Sep	Explore the data and make charts. Build features and the gap breakdown (Part B).	Figures ready for the slides; the feature table passes the "known before the election?" check.
Sat 26 Sep	Backtest the forecast models, forecast 2026, build profiles and the priority list.	The model beats "same as last time", or we say honestly that it doesn't.
Sun 27 Sep	Streamlit dashboard, slides with references, record the video with every member presenting.	Dashboard runs from a fresh start; video is 15 minutes or less.
Mon 28 Sep before 08:00	One member uploads the notebook, video and slides to NextCloud, creates a shareable link and emails it to Ms Boitshepo Sebushe (bsebushe@csir.co.za).	Link sent.

Sources for the figures quoted

ITWeb: IEC sees strong online voter registration uptake (74% online; 61% women; 50% aged 16–29).

<https://www.itweb.co.za/article/iec-sees-strong-online-voter-registration-uptake/xnklOqz1QrGM4Ymz>

GroundUp: Young people stay home on voters day (turnout among 20–29-year-olds: 50% in 2016, 35% in 2021).

<https://groundup.org.za/article/young-people-stay-home-on-voters-day/>

Joburg ETC: IEC records 1.7 million registration transactions.

<https://www.joburgetc.com/news/iec-records-1-7-million-transactions-voters-roll-grows-2-1-million/>

Independent on Saturday: youth registrations by province. <https://independentonsaturday.co.za/news/2026-08-04-iec-reports-significant-youth-turnout-in-voter-registrations-for-upcoming-local-government-elections/>

IEC Voter Registration Statistics: <https://www.elections.org.za/pw/StatsData/Voter-Registration-Statistics>

IEC municipal election results: <https://results.elections.org.za/home/downloads/me-results>